

```

#include <iostream>
#include <vector>
#include <algorithm>

class TrainSolver {
public:
    TrainSolver(int numCases) : t(numCases) {}

    void readData() {
        for (int i = 0; i < t; ++i) {
            int numTrains;
            std::cin >> numTrains;
            std::vector<int> trains(numTrains);
            for (int j = 0; j < numTrains; ++j) {
                std::cin >> trains[j];
            }
            trainData.push_back(trains);
        }
    }

    void solveAll() {
        for (const auto& trains : trainData) {
            solve(trains);
        }
    }

private:
    int t;
    std::vector<std::vector<int>> trainData;

    void solve(const std::vector<int>& trains) {
        int n = trains.size();
        std::vector<int> dp1(n, 1), dp2(n, 1);
        int ans = 0;

        for (int i = n - 1; i >= 0; --i) {
            for (int j = n - 1; j > i; --j) {
                if (trains[i] > trains[j])
                    dp1[i] = std::max(dp1[i], dp1[j] + 1);
                else
                    dp2[i] = std::max(dp2[i], dp2[j] + 1);
            }
            ans = std::max(ans, dp1[i] + dp2[i] - 1);
        }

        std::cout << ans << std::endl;
    }
};

```

```
int main() {  
    int t;  
    std::cin >> t;  
    TrainSolver solver(t);  
    solver.readData();  
    solver.solveAll();  
    return 0;  
}
```